

KFAE – FAE with First-Class Continuations

1 INTRODUCTION

KFAE is a toy language for the [COSE212](#) course at Korea University. KFAE stands for the [KFAE](#) language with the **first-class continuations**, and it supports the following features:

- **number (integer) values** (0, 1, -1, 2, -2, 3, -3, ...)
- **arithmetic operators**: addition (+) and multiplication (*)
- **immutable variable definitions** (val)
- **first-class functions** (=>)
- **first-class continuations** (vcc)

This document is the specification of KFAE. First, Section 2 describes the concrete syntax, and Section 3 describes the abstract syntax with the desugaring rules. Then, Section 4 describes the small-step operational semantics of KFAE.

2 CONCRETE SYNTAX

The concrete syntax of KFAE is written in a variant of the extended Backus–Naur form (EBNF). The notation `<nt>` denotes a nonterminal, and `"t"` denotes a terminal. We use `?` to denote an optional element and `+` (or `*`) to denote one or more (or zero or more) repetitions of the preceding element. We use **butnot** to denote a set difference to exclude some strings from a producible set of strings. We omit some obvious terminals using the ellipsis (`...`) notation.

```
// basic elements
<digit>    ::= "0" | "1" | "2" | ... | "9"
<number>   ::= "-"? <digit>+
<alphabet> ::= "A" | "B" | "C" | ... | "Z" | "a" | "b" | "c" | ... | "z"
<idstart>  ::= <alphabet> | "_"
<idcont>   ::= <alphabet> | "_" | <digit>
<keyword>  ::= "val" | "vcc"
<id>       ::= <idstart> <idcont>* butnot <keyword>

// expressions
<expr> ::= <number> | <expr> "+" <expr> | <expr> "*" <expr> | <id>
        | "(" <expr> ")" | "{" <expr> "}"
        | <id> "=>" <expr> | <expr> "(" <expr> ")"
        | "val" <id> "=" <expr> ";" <expr>
        | "vcc" <id> ";" <expr>
```

The precedence and associativity of operators are defined as follows:

Operator	Associativity	Precedence
*	left	2
+	left	1

3 ABSTRACT SYNTAX

The abstract syntax of KFAE is defined as follows:

Expressions	$\mathbb{E} \ni e ::= n$	(Num)		
	$e + e$	(Add)		
	$e * e$	(Mul)		
	x	(Id)	where	Numbers $n \in \mathbb{Z}$ (BigInt)
	$\lambda x.e$	(Fun)		Identifiers $x \in \mathbb{X}$ (String)
	$e(e)$	(App)		
	$\text{vcc } x; e$	(Vcc)		

The semantics of the remaining cases are defined with the following desugaring rules:

$$\mathcal{D}[\llbracket \text{val } x = e_1; e_2 \rrbracket] = (\lambda x. \mathcal{D}[\llbracket e_2 \rrbracket])(\mathcal{D}[\llbracket e_1 \rrbracket])$$

The omitted cases recursively apply the desugaring rule to sub-expressions.

4 SEMANTICS

We use the following notations in the semantics:

Values	$\mathbb{V} \ni v ::= n$	(NumV)
	$\langle \lambda x.e, \sigma \rangle$	(CloV)
	$\langle \kappa \parallel s \rangle$	(ContV)
Environments	$\sigma \in \mathbb{X} \xrightarrow{\text{fin}} \mathbb{V}$	(Env)
Continuations	$\mathbb{K} \ni \kappa ::= \square$	(EmptyK)
	$(\sigma \vdash e) :: \kappa$	(EvalK)
	$(+) :: \kappa$	(AddK)
	$(\times) :: \kappa$	(MulK)
	$(@) :: \kappa$	(AppK)
Value Stacks	$\mathbb{S} \ni s ::= \blacksquare \mid v :: s$	(List[Value])

The small-step operational semantics of KFAE is defined as follows:

	$\langle \kappa \parallel s \rangle \rightarrow \langle \kappa \parallel s \rangle$	
Num	$\langle (\sigma \vdash n) :: \kappa \parallel s \rangle$	$\rightarrow \langle \kappa \parallel n :: s \rangle$
Add ₁	$\langle (\sigma \vdash e_1 + e_2) :: \kappa \parallel s \rangle$	$\rightarrow \langle (\sigma \vdash e_1) :: (\sigma \vdash e_2) :: (+) :: \kappa \parallel s \rangle$
Add ₂	$\langle (+) :: \kappa \parallel n_2 :: n_1 :: s \rangle$	$\rightarrow \langle \kappa \parallel (n_1 + n_2) :: s \rangle$
Mul ₁	$\langle (\sigma \vdash e_1 * e_2) :: \kappa \parallel s \rangle$	$\rightarrow \langle (\sigma \vdash e_1) :: (\sigma \vdash e_2) :: (\times) :: \kappa \parallel s \rangle$
Mul ₂	$\langle (\times) :: \kappa \parallel n_2 :: n_1 :: s \rangle$	$\rightarrow \langle \kappa \parallel (n_1 \times n_2) :: s \rangle$
Id	$\langle (\sigma \vdash x) :: \kappa \parallel s \rangle$	$\rightarrow \langle \kappa \parallel \sigma(x) :: s \rangle$
Fun	$\langle (\sigma \vdash \lambda x.e) :: \kappa \parallel s \rangle$	$\rightarrow \langle \kappa \parallel \langle \lambda x.e, \sigma \rangle :: s \rangle$
App ₁	$\langle (\sigma \vdash e_1(e_2)) :: \kappa \parallel s \rangle$	$\rightarrow \langle (\sigma \vdash e_1) :: (\sigma \vdash e_2) :: (@) :: \kappa \parallel s \rangle$
App _{2, \lambda}	$\langle (@) :: \kappa \parallel v_2 :: \langle \lambda x.e, \sigma \rangle :: s \rangle$	$\rightarrow \langle (\sigma[x \mapsto v_2] \vdash e) :: \kappa \parallel s \rangle$
App _{2, \kappa}	$\langle (@) :: \kappa \parallel v_2 :: \langle \kappa' \parallel s' \rangle :: s \rangle$	$\rightarrow \langle \kappa' \parallel v_2 :: s' \rangle$
Vcc	$\langle (\sigma \vdash \text{vcc } x; e) :: \kappa \parallel s \rangle$	$\rightarrow \langle (\sigma[x \mapsto \langle \kappa \parallel s \rangle] \vdash e) :: \kappa \parallel s \rangle$

where $\rightarrow^*$ is the reflexive-transitive closure of $\rightarrow$ and denotes the repeated application of $\rightarrow$:

$$\begin{array}{c} \langle \kappa \parallel s \rangle \rightarrow^* \langle \kappa \parallel s \rangle \\[10pt] \frac{\langle \kappa \parallel s \rangle \rightarrow^* \langle \kappa' \parallel s' \rangle \quad \langle \kappa' \parallel s' \rangle \rightarrow \langle \kappa'' \parallel s'' \rangle}{\langle \kappa \parallel s \rangle \rightarrow^* \langle \kappa'' \parallel s'' \rangle} \end{array}$$

The evaluation result of an expression e is the value v if

$$\langle (\emptyset \vdash e) :: \square \parallel \blacksquare \rangle \rightarrow^* \langle \square \parallel v :: \blacksquare \rangle$$